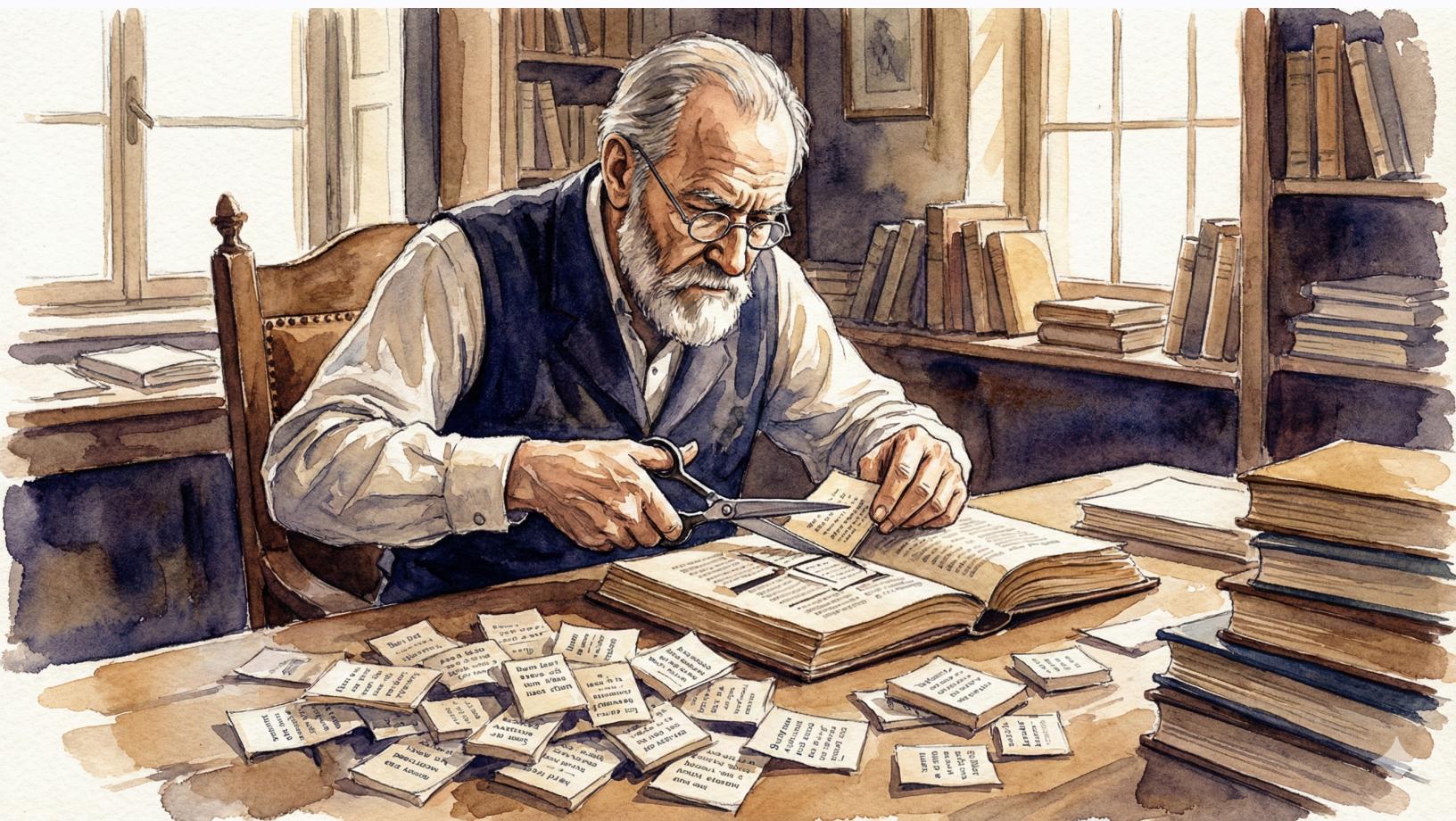


Week 3 Lesson Plan

Enterprise RAG: Measuring the Size of Thoughts — To Chunk, or Not to Chunk?

ASIF QAMAR



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First draft, June 20, 2026

Measuring the Size of Thoughts

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Not everything that can be counted counts, and not everything that counts can be counted.

— William Bruce Cameron

The beginning of wisdom is the definition of terms.

— Socrates (attributed)

If you want the Day-1 and Day-2 versions of this recap — the Library of Alexandria, the polar bear in the equatorial band, the grin without a cat, the force-field reading of contrastive training — they live in Chapters ?? and ??. I will not re-derive them here. I will *use* them.

A note on what we are deferring. We still owe ourselves the detailed mathematics of *contrastive loss* — triplet, InfoNCE, the temperature term, the attract-repel force field. I am moving it to the back half of the evening, or to next week, deliberately. First, I want your hands dirty on the capstone before the night is out. Second, we have just written a careful tutorial on contrastive objectives (`loss_survey_tufte.tex`); read it as self-study so you arrive next week ready to *discuss* rather than *discover*. Third, the TAs have already started you on contrastive-loss lab code with a short primer — so the ideas are not idle while we wait.

Capstone, today. We also open the floor this session. Each team will present the plan for the capstone it has selected: the corpus, the users and their questions, the retrieval challenge that makes the corpus hard, and the metric by which you will know you have succeeded. Bring it in rough form — a sketch we can pressure-test together is worth more than a polished slide. Notice how directly today’s topic bears on your plan: the *shape* of your documents (prose? tables? scanned pages?) already constrains how you should decompose them.

A Gentle Re-Entry

TWICE NOW WE HAVE STOOD AT THE EDGE OF A SPHERE. On Day 1 we argued that meaning has become geometry: a modern machine does not reason in words but in *vectors*, points on the surface of a high-dimensional globe, where nearness is kinship and direction is meaning. We met the curse of dimensionality and its merciful twin, the concentration of measure, and we watched a retriever fetch the right passage about *friendship with animals* even though the winning passage contained none of those words. On Day 2 — “The Shape of a Decision” — we earned that geometry. We built, almost from nothing, the softmax and the attention mechanism; we saw surprise as the negative log-likelihood; and we ended on *anisotropy*, the quiet tendency of embeddings to huddle in a narrow cone rather than spread honestly across the hypersphere.

This week we take the next, deceptively simple step. Before a single vector can be computed, a document must be turned into the *things* we embed. That act has a name — **chunking** — and it is treated everywhere as settled plumbing: parse the document, slice it into passages, embed, retrieve. Our whole week is an argument that this moment deserves to be slowed down, because it hides the first irreversible decision in the entire pipeline.

HERE IS THE SINGLE THREAD. A document is not one thought; it is an *ensemble* of thoughts. To embed it well, we must cut it into units that are each, as nearly as possible, a single coherent thought. But every cut destroys something — a pronoun’s antecedent, a qualifier’s claim, a figure’s caption. So the real question of the week is not “what is the best chunk size?” It is two deeper questions, asked in order: *should we chunk at all?* And if we must, *how do we lose wisely?*

What Must You Carry Forward

WHEN TODAY’S SESSION CONCLUDES, you should be able to do each of the following without hand-waving:

- Explain why **every embedding is a lossy projection**, and why the *first* projection — from source to embedded unit — is the one no downstream cleverness can undo.
- State the **mixing principle**: why embedding two orthogonal thoughts

as one vector produces a point that is “about” neither, and where that point lands in the geometry of Day 1.

- Give the three engineering reasons we chunk *anyway* — context windows, retrieval precision, and cost — and rebut the claim that long context windows have made chunking obsolete.
- Diagnose at least six **pathologies of chunking** by name — broken endophora, severed discourse relations, the negation/scope trap, dissolved lexical cohesion, lost bridging inference, and the centroid delusion — and recognize each in a real failure.
- Distinguish **fixed-size**, **recursive**, **semantic**, and **hierarchical** chunking, and say precisely what each does and does not fix.
- Explain **contextual chunking** and **late chunking**, and articulate the difference between repairing in *text space* versus in *latent space*.
- Make the case — with evidence, not slogans — for **not chunking at all**: page-image retrieval with ColPali-style late interaction, and when a tournament would pick it over parse-and-chunk.
- Design a **representation tournament** for your own capstone corpus: the competitors, the held-out queries, and the metric that decides.

If you can do these, you are ready for next week, where we will argue that the wisest move of all is to *transform* the source before you ever cut it.

The Central Provocation: To Chunk, or Not to Chunk?

NEARLY EVERY TUTORIAL ON RAG BEGINS THE SAME WAY: fetch a document, parse it, chunk it, embed the chunks, retrieve. The sequence is so universal that questioning it feels almost heretical. Yet heresy, grounded in evidence, is the birthplace of progress. Before we ask *how* to chunk, we must ask whether chunking is the right abstraction at all. Recent advances in vision-language models (VLMs) have unsettled the orthodoxy. For whole classes of documents — the ones rich in tables, charts, figures, and layout — embedding a *page as an image* can outperform a carefully engineered parse-and-chunk pipeline. This is not an anomaly; it is a signal. The correct stance, therefore, is not dogma but empiricism: competing representations are placed in a tournament, and the only arbiter is measurement.

The dominant narrative is **parse** → **chunk** → **embed** → **retrieve**. A modern survey of RAG catalogues dozens of pipeline variations — yet almost all of them open with the same unquestioned first step. We shall question it, beginning with parsing itself.

The Six Mental Frames of the Week

Hold these six frames; everything today is a variation on them.

- **Chunking is not preprocessing.** It is an *ontological* decision about what constitutes a thought.
- **Every embedding is a lossy projection.** The first projection into latent space is irreversible.
- **There is no universally optimal decomposition.** Document structure, modality, and task jointly determine what works.
- **Representation choice is experimental.** Competing approaches must be run as tournaments, not presumed.
- **Small chunks trade purity for amnesia; large chunks trade context for entanglement.**
- **Re-contextualization belongs downstream.** Context is best restored at retrieval or generation time, not baked blindly into the embedding.

Act I: The First Cut Is the Deepest

ACT I ESTABLISHES THE FOUNDATIONS: why the first projection is irreversible, why we chunk despite that, and why the innocent question “how big should a chunk be?” has no universal answer.

The First Projection Is Irreversible

BEFORE ANY MECHANICS, internalize a principle that governs everything that follows: every embedding is a lossy projection, and the first projection from source into latent space cannot be undone. What the encoder cannot see, the generator cannot recover.

Remember the children’s telephone game from Day 1 — the whisper that degrades with each relay. Embedding is the first relay. If the very first step introduces significant loss, no amount of downstream sophistication — no reranker, no clever prompt — can restore it. This is why chunking is not preprocessing. It is the *first ontological decision* in your retrieval pipeline: what you decide to embed as one unit fixes what the system can and cannot retrieve.

Cautionary tale. An engineer who spends weeks tuning rerankers and prompts while leaving chunking at the default 512-token setting is polishing the upper floors of a building whose foundation is cracked. The greatest leverage in most RAG systems is not the generator; it is the quality of the input to the retriever — and chunking determines that quality.

The Information Bottleneck

This loss is structural, not incidental. The **information bottleneck** frames representation learning as compression: given input X and target Y , find a representation Z that keeps maximal information about Y while discarding the rest of X ,

$$\mathcal{L}_{\text{IB}} = I(X; Z) - \beta I(Z; Y),$$

where $I(\cdot; \cdot)$ is mutual information and β tunes the trade-off. When we embed a chunk, we pass it through exactly such a bottleneck. If the chunk holds two distinct thoughts — cows *and* the stock market — the bottleneck must crush both into one point. The result is faithful to neither. This is why chunk *composition* matters so profoundly.

Why We Chunk: Atoms of Thought

SUPPOSE WE HAVE DECIDED TO PARSE AND CHUNK. Why chunk at all — why not embed whole documents? The answer flows from first principles. A machine maps text to a point on a manifold, and that mapping is meaningful only when each point represents one *semantic unit* — one atom of thought. A document is, by definition, not a single unit; it is a collection of them. Chapters are themes, sections are topics, subsections are smaller units. The author’s act of dividing a long document is itself explicit signal: *here a context switch occurs; a new thought begins*.

The Mixing Principle

Write “the cow jumped over the moon.” One atomic thought. Now write “Nvidia stock collapsed today.” A second thought, orthogonal to the first. In a thousand-dimensional space these should land far apart. But concatenate them into one embedding and you create a composite point representing a thought that does not exist. In the concentration-of-measure framework from Day 1, that composite lands in the *equatorial band* where noise lives — equidistant from both clusters. Queried about cows, it is “kind of about cows”; queried about stocks, it shows up too. It is noise on both. A single fixed-dimensional vector simply cannot faithfully encode two orthogonal thoughts; the channel capacity is finite.

There is also a practical argument. Chunking enables *selective retrieval*. A monolithic document vector forces the retriever into an all-or-nothing verdict; it cannot say “Chapter 3 is relevant but Chapters 5 and 7 are not.” Breaking the document into chunks lets the system surface exactly the passages a query needs, with minimal surrounding noise. And there is an economic argument. A long context incurs quadratic

cost in the attention mechanism: a thousand tokens cost on the order of 10^6 operations; a million tokens, 10^{12} . By retrieving only the relevant fraction, RAG keeps the context window small and the cost low. Attempts to dodge this with ever-longer windows hit diminishing returns: sparse-attention models such as Longformer and BigBird shave the constant but not the fundamental $O(n^2)$ bound, and even when a model *can* ingest a long context, it suffers the “lost in the middle” effect — information buried mid-context is attended to least reliably. Selective retrieval keeps the relevant content front and center.

The Boundary Problem and the 512-Token Default

THE FIRST PATHOLOGY IS DECEPTIVELY SIMPLE: where do we draw the line? A document is a continuous stream of meaning, and the question of where one chunk ends has no ground truth. Too small, and we lose context; too large, and we conflate distinct thoughts. There is no universal right answer — optimal granularity depends on query distribution, domain density, and downstream task.

The 512-Token Default

Fixed-size chunking is the worst offender, and 512 tokens is the most common default. Let us reckon: a line holds 8–10 words, roughly 10–13 tokens; a page holds 30–40 lines, hence ≈ 300 –400 tokens. So a 512-token chunk is about one page. What is the probability that an arbitrary page contains exactly one coherent thought? For a dense newspaper article, very low. For a discursive academic — a friend once left my lecture for coffee, returned an hour later, and exclaimed, “You are *still* on the same point!” — also very low. The 512-token default is not a considered choice; it is a baseline for expedience. Yet adopted unreflectively it cascades through the entire pipeline, silently fixing what the system can ever retrieve.

This is the first lesson of chunking: question the orthodoxy. Do not be indoctrinated into the unthinking sequence every tutorial prescribes. Think from first principles; understand what you gain and lose at each step.

Act II: A Field Guide to What Chunking Destroys

ACT II IS A FIELD GUIDE TO FAILURE. When a RAG system returns a confidently wrong answer, the root cause is rarely a bad embedding model or a poor prompt. More often it is a chunking decision made months earlier, silently severing a connection the system can never recover. Naming these pathologies gives you a *diagnostic vocabulary* — so that when your system fails, you can ask precisely *which* kind of failure it is.

Endophora and Broken References

LANGUAGE REFERS TO ITSELF. “The cow jumped over the moon. It was rather pleased with itself.” The pronoun “it” points backward to “cow” — this is *anaphora*. Cut the two sentences into different chunks and the second becomes incoherent. The mirror image is *cataphora*, a forward reference: “He had no money. Yet John hoped to eat anyway.” Both are forms of **endophora** — text referring to other text in the same document — and both saturate technical and legal writing. “As we saw in Equation 3.6...” is meaningless if Equation 3.6 sits three chapters away in another chunk.

This is the purity–amnesia trade-off in miniature: **small chunks trade purity for amnesia**. A fifty-token chunk likely holds one thought, but may lack the context to resolve its pronouns and abbreviations. In Indian philosophy all meaning is said to live in one syllable — *Om* — but that is meaning compressed by millennia of shared context, which no embedding model possesses. In practice, meaning without context is noise.

The nuqta. In Urdu calligraphy a single dot — the *nuqta* — can invert a word: move it and *Khuda* (God) becomes *Judaa* (parted, gone). Punctuation and stop words play this role in prose: a comma turns a list into a sequence; a “not” inverts a truth; a “however” reverses a conclusion. Sever these marks from the clauses they govern and the fragment may read fluently while meaning the opposite of what the author intended.

Discourse Structure and the Severing of Relations

DOCUMENTS ARE NOT SEQUENCES OF INDEPENDENT SENTENCES. They are hierarchical networks of *rhetorical relations*. Rhetorical Structure Theory formalizes what you already feel: a document decomposes into a tree of units bound by relations — cause–effect, contrast, elaboration, evidence, concession. These are the logical skeleton of the argument. Sever them and you do not merely lose information; you get *structurally misleading* information that reads true but is not.

Consider:

“The drug reduced tumor volume by 40% ($p < 0.01$). However, the placebo group showed a 15% reduction, suggesting a placebo effect. Adjusted for this baseline, the drug’s net effect was a modest 25%, which did not reach clinical significance.”

Stephen Toulmin models an argument in six parts: **claim**, **data**, **warrant**, **backing**, **qualifier**, and **rebuttal**. Chunking can sever any link. A rebuttal without its claim becomes a mysterious negative; a claim without its qualifier becomes absolute; a warrant without its backing becomes an unsupported assumption. When a retriever surfaces a chunk, ask: is this a complete Toulmin unit?

Three units bound by a *concession* and an *evaluation*. Surface only the first sentence — the one a query about efficacy would rank highest — and you deliver the *opposite* of the document's conclusion. The chunk is not incomplete; it is actively deceptive.

Lawyers call this “taking out of context”; engineers call it a chunking failure. They are the same thing. A clause reading “The licensee shall have no obligation to pay royalties” is devastating — until you read the preceding condition, “In the event that cumulative sales do not exceed \$1 million in any calendar year.” The conditional frame *is* the meaning.

The Negation Trap and Scope Failures

SCOPE FAILURES DESERVE THE MOST ATTENTION because they do not merely lose information — they *invert* it. Negation is the most dangerous culprit. “The meta-analysis found *no* significant evidence that Compound X increases cardiac risk in patients under 65. However, in patients 65–80 with hypertension, the data suggested a concerning elevation.” Surface only the second sentence and the reassurance-with-a-caveat becomes pure alarm; surface only the first and the caveat vanishes. Neither is the truth. The truth is the *relation* between them — and that relation lives exactly at the boundary where your chunker cuts.

The same trap springs on *conditional* scope (“If the debt-to-income ratio exceeds 43%, the following requirements apply...” — chunk away the condition and the requirements appear universal), on *counterfactual* mood (“Had the rate stayed at 3%, the portfolio would have appreciated 12%” — the subjunctive that signals non-reality is the first casualty), and on *epistemic hedging*.

Presupposition Accommodation

When a paper writes “we tentatively suggest, on preliminary evidence, that mechanism Y *may* play a role,” the hedge scopes the whole claim. Three sentences later the authors use the shorthand of an established fact, as authors do. If the elaboration lands in a different chunk from the hedge, a tentative hypothesis is silently promoted to certainty. Epistemologists call this **presupposition accommodation**: encountering a claim without its epistemic frame, the reader (or retriever) treats it as shared background. This is the most insidious chunking failure — the one that produces no error signal at all. The system is confidently, silently wrong.

Lexical Cohesion and Definition–Use Chains

DOCUMENTS ALSO COHERE THROUGH *implicit* CONNECTIONS. Halliday and Hasan identified five mechanisms of textual cohesion; the subtlest is *lexical cohesion* — the cumulative effect of related words that build a sustained semantic field, an *isotopy*. A legal brief that threads “negligence,” “duty of care,” “breach,” “proximate cause,” and “damages” across its pages makes each term resonate with the others. Chunk the brief and each fragment carries only a few of these terms; the field effect dissolves.

This matters geometrically. A chunk holding “breach” and “proximate cause” embeds firmly in legal territory. A chunk holding “breach” alone could mean a breached wall, a breach of protocol, or a whale breaching the surface. The isotopy disambiguates; chunking destroys it; retrieval suffers. This explains a genuinely puzzling production observation: retrieval quality sometimes *degrades* as chunks get smaller, even though smaller chunks should be more focused. The focused chunk has lost its lexical company.

Bridging Inference and Genre Blindness

THERE IS A CONNECTION SUBTLER STILL, one with nothing explicit to detect. “John walked into the room. The chandelier was magnificent.” No pronoun, no connective — yet every reader knows the chandelier is *in* the room. The definite article signals that the chandelier belongs to the *frame* that “room” activated. This is **bridging inference**, and Schank’s theory of *scripts* explains its reach: “the restaurant” activates waiter, menu, bill, tip, so “she left a generous tip” is coherent only while the restaurant script is live. Chunk it away and the embedding captures the act of tipping but not its situation. In technical text the danger sharpens: “the potential” (gravitational? electric?), “the spread” (bid-ask? credit? yield?) — the surrounding text is the only disambiguator, and chunking severs it invisibly.

A STRUCTURAL FAILURE PRECEDES ALL OF THESE: **genre blindness** — treating a legal contract, a novel, and a scientific paper as if they shared one information structure. The contract is dense (every clause carries force); the novel is sparse (pages of atmosphere around one pivotal line); the paper has rigid implicit structure (abstract, methods, results, each presupposing the others). Your chunker gives all three 512 tokens. And density varies *within* a document too: an abstract is extraordinarily

Definition–use chains are the same wound in sharper form. A contract defines “Covered Losses” once, in §2.1; thereafter every use carries that definition’s weight. Chunk the document and a §7.4 passage about “Covered Losses” reaches the retriever orphaned from its definition — two generic English words where a precise legal term was meant. The same pattern haunts API docs, medical acronyms, and standards documents: one definition, many uses, and chunking preserves the uses while severing the source.

dense, an introduction discursive. Optimal chunking would adapt its granularity to this variation — yet almost no production system does, because measuring density in real time requires the very understanding the system is trying to achieve. Tone is the final casualty: Swift’s “A Modest Proposal” is satire only in the contrast between earnest setup and monstrous punchline; chunk them apart and one reads as policy, the other as lunacy, and neither is the truth.

Pathologies in Embedding Space

THE PATHOLOGIES SO FAR ARE TEXTUAL. But chunking also distorts the *embedding space* itself. The most important is the **centroid delusion**. A chunk’s embedding is the centroid of its content — the *average* meaning. Pack three distinct thoughts into a 400-token chunk and its embedding sits equidistant from all three, which is to say near none of them. It is a phantom: a location where no actual thought resides. You are hunting a needle, and the embedding has helpfully averaged it with the haystack. This is Simpson’s paradox in vector form: an aggregate can be far from *every* constituent.

Query–chunk asymmetry. Queries are short and sharp; chunks are long and diffuse. A focused query point matched against a blurred chunk centroid *underestimates* true relevance, so the system is biased toward homogeneous chunks and against heterogeneous ones — regardless of which holds the answer. It is retrieving the most internally coherent chunks, not the most relevant ones. Those are not the same thing.

Anisotropy — our Day-2 antagonist — compounds this. In an anisotropic space, large diverse chunks drift toward the mean direction of the cone and become similar to *everything* yet *particularly* similar to nothing: the “popular but useless” results that appear in every query’s top-*k* and answer none of them. The cure is not filtering but re-chunking into focused units whose embeddings escape the dense center.

When the Embedder Cannot See the Comma

Take Lynne Truss’s title *Eats, Shoots & Leaves*. With commas it is a gunslinger; without, a panda. Now test three sentences — (1) “Eats, shoots, and leaves,” (2) “Eats shoots and leaves,” (3) “A panda eats bamboo shoots and leaves.” Any reader knows (2) is closer to (3) than to (1). In our classroom experiments, many popular encoders get the ordering *wrong*: they weight lexical overlap above the compositional semantics that punctuation controls. The embedder cannot see the comma’s work. The lesson: if your chunks differ by syntactic cues rather than vocabulary — as legal, medical, and regulatory text so often do — your embedding model may simply fail to tell them apart. Embedding model selection is itself a first-order chunking decision.

AND YET, CHUNKING LIVES ON. After that catalogue of horrors, why chunk at all? Because chunking is not born of ignorance; it is a *compromise* born of constraints. **Context windows** are finite: ten thousand

fifty-page documents will not fit, however generous the window, so you must select units smaller than documents. **Retrieval precision** demands it: whole-document embeddings are diffuse averages that may never crack the top- k , and a document never retrieved is useless. **Cost** demands it: processing a fifty-page unit is orders of magnitude more expensive than fifty one-page chunks. And **answer isolation** rewards it: a tight chunk gives the model less hay to search for the needle. A lossy signal that reaches the model beats a lossless one buried on page thirty-seven of a document ranked eleventh. So chunk — boldly, thoughtfully, with your eyes open — and make sure what survives is enough.

Act III: Two Roads — Better Chunking, or No Chunking?

ACT III OFFERS TWO ROADS OUT OF THE SWAMP. The first improves the cut: a spectrum of strategies that lose less. The second questions the cut entirely: skip parsing and chunking, and retrieve the page as an image. We walk both, then let measurement decide.

Road One, Part A: A Taxonomy of Chunking Strategies

FIXED-SIZE CHUNKING is the baker's approach: treat text as a loaf and cut equal slices. Simple, deterministic, fast — and wrong for text, because a fixed boundary may bisect a thought, a sentence, even a word. What is worse than a worm in your apple? Half a worm. **Recursive chunking** improves matters by cutting preferentially at word, then sentence, then paragraph boundaries — but it is still about *size*, not meaning, and sentence boundaries barely exist in Chinese or Japanese while paragraph boundaries swing wildly between a journalist and Sir Francis Bacon.

SEMANTIC CHUNKING cuts where meaning actually shifts. Embed consecutive sentences, measure the distance between neighbors, and cut where the distance spikes — a genuine topic transition. Libraries such as **Chonkie** implement this. It is far better than fixed-size, at a modest cost in compute. But it improves *where* we cut; it does not heal what the cut severs. “It was rather pleased with itself” is correctly marked as a unit's end — and remains incomplete without its referent.

HIERARCHICAL CHUNKING embraces that documents have levels. Chunk coarsely (sections, ≈ 300 – 500 tokens) and finely (atoms, ≈ 50 – 80 tokens), building a tree of parents and children. Parents preserve context; children are coherent. At retrieval time the system chooses which to feed — and that choice is a trap.

Two Traps of Hierarchical Retrieval

Trap 1: the illusion of overlap. Overlapping windows ensure a severed sentence survives intact in some neighbor, but do nothing for references spanning paragraphs or pages, and they breed near-duplicate chunks that crowd the top- k . Overlap is a local bandage on a global wound. **Trap 2: the derailment of parents.** Substitute a parent for richer context and the LLM receives every sentence in it — including the irrelevant ones — with no way to know which triggered retrieval. A parent about supply-chain logistics that happens to mention pricing will weave logistics into your pricing answer. The parent does not merely add noise; it *derails* generation.

Re-Contextualization: Contextual and Late Chunking

RATHER THAN INFLATE CHUNKS, RESTORE CONTEXT DYNAMICALLY. **Contextual chunking** repairs in *text space*: chunk small, then for each chunk C_i use an LLM to read the whole document and prepend the minimal context that completes it, embedding $C'_i = \text{CTX}_i \parallel C_i$. The chunk “It has a population of 3.6 million” becomes “Berlin is the most populous city of Germany. It has a population of 3.6 million.” Described in Anthropic’s widely cited note on contextual retrieval and studied by Das et al., it elegantly addresses endophora — at the cost of one LLM call per chunk, tamed in practice by a local model and prompt caching.

LATE CHUNKING repairs in *latent space*, and it is the more elegant idea. In the villages of North India, grandmothers cut old saris into patches and stitch them into a *razai* — a quilt that holds every memory in one warm whole. Late chunking runs that in reverse: let the encoder stitch the whole document together — every token absorbing every other through attention — and only *then* cut. If the encoder produces contextualized latents $h_i = \text{encode}(t_i \mid t_1, \dots, t_n)$, each h_i depending on *all* tokens, then

$$v_{\text{chunk}} = \text{MeanPool}(h_{i_1}, h_{i_1+1}, \dots, h_{i_2}),$$

where i_1, i_2 are the chosen boundaries. Compare traditional chunking,

which first extracts text and embeds it *without* its surroundings, $v'_{\text{chunk}} = \text{embed}(t_{i_1:i_2})$. The difference is the difference between a word that knows its neighbors and a word that has been orphaned. A chunk whose text reads only “it jumped over the moon” can still be retrieved by “what did the cow do?” — because when “it” was encoded, it soaked up “cow” from prior context. It feels like magic; it is geometry.

A Practical Pipeline

IN PRACTICE THE BEST SYSTEMS COMBINE THESE AS LAYERS OF DEFENSE:

(1) **markup chunking** first — if the document has headings, sections, or markdown structure, respect it; the author already marked the boundaries. (2) **semantic hierarchical chunking** with a tool like Chonkie, coarse then fine, on meaning rather than size. (3) **contextual chunking** if budget allows, prepending inferred context to each fine chunk. (4) **late chunking** before embedding, slicing contextualized latents at the chosen boundaries. (5) **store** in a vector database with metadata — section, position, parentage — intact.

This pipeline is a framework, not dogma. For any new domain, run a tournament: try strategies, measure retrieval quality, and let the empirical winner decide. No single technique solves all problems — markup respects intent, semantic finds boundaries, contextual recovers references, late chunking integrates global context. *All chunking is wrong, but some is useful*: the goal is not to eliminate loss but to choose which losses you can afford.

Road Two: Are We Barking Up the Wrong Tree?

EVERY STRATEGY ABOVE SHARES A HIDDEN PREMISE: that we should parse the document into text at all. Ask what parsing destroys. Picture a single page of a physics textbook: running prose, the equation $PV = nRT$, a pressure–temperature graph, a table of measurements, a diagram of a piston with steam rising. A parser extracts “As shown in the figure above, pressure and temperature are proportional. $PV = nRT$ formalizes this.” What is lost? The spatial relationships, the diagram, the graph, the tabular data — and the referential integrity, for “as shown in the figure above” now floats untethered.

Both methods attack endophora, but differently. Contextual chunking is more *interpretable* — you can read the prepended context and check it. Late chunking is more *elegant and efficient* — chunking and embedding merge into one pass over a local encoder. For mission-critical corpora, do both: contextualize the text, then late-chunk the result.

A good leader, it is said, holds both an **earthworm’s view** — local, granular, every grain of soil — and an **eagle’s view** — global, panoramic, the whole landscape. Small contextualized chunks give you the earthworm; hierarchical and late chunking give you the eagle. A well-designed pipeline gives you both.

Exophoric Reference

Recall the dosa-shop directions from Day 1. The American directions — “Take 101, exit Page Mill, turn left” — are already a flat sequence; they survive parsing. The Indian directions — “past the dosa shop, skip that one, the next dosa shop, right at the cobbler, look for the peepal tree” — encode meaning through *spatial relationships between landmarks*. Parsing flattens everything into the American form, destroying exactly the **exophoric** texture that layout carries. A printed page is a two-dimensional signal; parsed text is one-dimensional. Flattening 2D to 1D is not a minor inconvenience — it is a fundamental reduction in channel capacity.

VISION-LANGUAGE MODELS INVERT THE PREMISE. In a striking result, Alibaba Research took scientific documents and, instead of parsing, rendered each page to an *image*, embedded the screenshots with a VLM, and indexed those. Semantic search over page images dramatically outperformed parse-and-chunk. The intuition echoes Day 1: an image is high-bandwidth: a photograph of an elephant conveys texture, proportion, and presence that no caption matches. VLMs often handle visual information *better* than text because they preserve the spatial grammar — table position-as-semantics, the numerator above the denominator, indentation that carries legal force — that parsing flattens away. How? The Vision Transformer cuts a page into a grid of 16×16 -pixel patches and treats each as a token; a 1024×1024 page becomes ≈ 4096 patch tokens, and self-attention discovers relationships between any two regions — caption and figure, footnote marker and text. Crucially, this is the *same* self-attention as a language model: to the transformer, a patch and a subword are merely different tokens. That architectural indifference to modality is the foundation of the whole enterprise.

COLPALI MADE THIS PRACTICAL AT SCALE. It marries a vision encoder to ColBERT-style *late interaction*: rather than crush a page into one vector, it keeps a vector per patch and scores relevance by **MaxSim** — for each query token, find its most similar page patch, then sum. The system can, in effect, *point at the region of the page* that answers the query, without ever extracting text. And a text query can match an image patch because contrastive training (CLIP, SigLIP) projects both modalities into one shared space: the bridge is geometric, not symbolic.

Where the Field Stands in 2026

ColPali was not the end but the beginning of a family. The open **ColVision** line now spans ColQwen2.5, the smaller **ColSmol**, and the Qwen3-based **ColQwen3-4B**, which posts state-of-the-art numbers on **ViDoRe**, the standard visual-document-retrieval benchmark. The practical worry — multi-vector storage — is being engineered away: **HPC-ColPali** compresses patch embeddings via k -means quantization and dynamic pruning; training-free pooling toolkits and pyramid indexing shrink the index further; and **NanoVDR** distills a 2 B visual retriever into a 70 M text-only encoder — and **ColModernVBERT**, at ≈ 250 M parameters, comes within a fraction of a point of ColPali's NDCG@5 with roughly $10\times$ fewer parameters. The headline finding for *your* capstones: **financial PDFs** — dense with tables and figures — show the widest gap, on the order of $\sim 62\%$ dense recall for text-only versus $\sim 84\%$ for a ColQwen-class visual retriever. If your corpus looks like that, page-image retrieval is no longer exotic; it is the baseline you must beat.

INTELLECTUAL HONESTY demands the trade-offs. Vision retrieval gives up exact text fidelity (a VLM's reading is approximate — for character-perfect keyword search, parsed text is indispensable); it gives up cross-page attention (a 500-page PDF is 500 independently encoded images); and it costs more compute per page (though the gap narrows each hardware generation). It is precisely the first limitation — approximate text fidelity — that makes the next result so surprising.

DeepSeek-OCR: Contexts Optical Compression

DeepSeek asked the inverse of the usual OCR question: how few *vision* tokens does it take to faithfully *encode* a page of text? Their DeepEncoder — local-attention (SAM), a $16\times$ convolutional compressor, then global-attention (CLIP) — produces as few as 64 vision tokens per page, from which a compact decoder reconstructs the full text at 97% precision; even at $20\times$ compression, accuracy holds near 60%. The deeper message: if a decoder can rebuild 97% of a page from 64 vision tokens, those tokens preserve nearly all its semantic content. So we can take page images straight to vision tokens for retrieval, bypassing the entire parse-chunk-tokenize-embed pipeline — every stage of which is a lossy projection. The page image is already a remarkably efficient semantic container.

SO: PARSE AND CHUNK, OR NOT? It depends — and the honest engineer runs the experiment. If your documents are mostly prose — essays, narratives, transcripts — parse-and-chunk remains sensible. If they are technical, dense with diagrams and tables, a **tournament** is warranted: text parse-and-chunk versus direct page-image embedding versus a

hybrid index where both contribute and reinforce. The no-free-lunch theorem guarantees no single strategy dominates all domains. In our consulting work, running this tournament is the single highest-leverage thing a RAG team can do — and the winner often surprises everyone.

A Calibration Note

THIS SESSION WAS DENSE. We argued that the first projection is irreversible; we built the case for chunking from atoms of thought, context windows, precision, and cost; we toured a field guide of pathologies — endophora, discourse relations, scope inversion, lexical cohesion, bridging inference, genre blindness, and the centroid delusion; we climbed the taxonomy from fixed-size to semantic to hierarchical to contextual to late chunking; and we put the whole orthodoxy on trial against page-image retrieval and the 2026 ColPali family.

Hold the six frames. Chunking is not preprocessing but an ontological decision; every embedding is a lossy projection; there is no universally optimal decomposition; representation choice is experimental; small chunks trade purity for amnesia; re-contextualization belongs downstream. Carry these and you will never again accept a 512-token default without asking what it costs you.

Chunking is the first opportunity to lose information, and every step thereafter struggles to recover what was lost at the beginning. The goal is not to lose nothing — that is impossible. The goal is to lose wisely. And the wisest move of all may be to transform the source before you ever cut it.

Housekeeping. Two threads continue outside this chapter. First, *capstone plans*: each team presents its corpus, users, retrieval challenge, and success metric today — and after this session you should be able to name which decomposition strategy your corpus demands, and design the tournament that proves it. Second, *contrastive loss*: the detailed derivations are deferred so we can get your hands dirty sooner. Read `loss_survey_tufte.tex` as self-study, lean on the TAs' lab primer, and arrive next week ready to discuss.

NEXT WEEK we pull the thread we left dangling. Every technique today assumed you start from the raw source and cut. But the source was written for a *human* reader who fills gaps and reads between lines — not for your retriever. What if, before chunking, you *transformed* the source into something purer: reformulated, recontextualized, enriched for retrieval? The source is the ore; the *derivative artifact* is the refined metal — and it is the metal you should chunk. *Picture abhi baaki hai, mere dost* — the movie is not over. This is the intermission.

Essential Readings

THESE ARE REQUIRED. Each develops a thread we opened today. Skim broadly; read at least one deeply.

- **Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*.** The canonical RAG paper — what retrieval was meant to solve, and what it was not.
- **Gao et al., *Retrieval-Augmented Generation for Large Language Models: A Survey*.** A map of the design space: chunking strategies, retrieval choices, evaluation. Use it to place today’s ideas in context.
- **Faysse et al., *ColPali: Efficient Document Retrieval with Vision Language Models*.** The paper behind “Road Two” — late interaction over page patches, and the ViDoRe benchmark.
- **Anthropic, *Contextual Retrieval***, with Das et al.. Repairing endophora in text space; read these as the practical companion to the theory of broken references.
- **Jina/Cohere, *Late Chunking***, with Wang et al.. Repairing endophora in latent space — the most elegant remedy we covered.
- **Wei et al., *DeepSeek-OCR: Contexts Optical Compression*.** The 64-tokens-per-page result that reframes the page image as an efficient semantic container.

Oliver Twist’s List of Readings

OPTIONAL, BUT EACH ONE DEEPENS THE WEEK. For the voracious mind.

- **Khattab & Zaharia, *ColBERT*.** The late-interaction / MaxSim paradigm that ColPali extends into vision.
- **The ColVision family and ViDoRe.** The living state of the art — ColQwen3, ColSmol — and the benchmark that ranks them.
- **Storage-efficient visual retrieval:** HPC-ColPali, training-free pooling, pyramid indexing, and the NanoVDR distillation. Read these if multi-vector storage is what is stopping you from shipping page-image retrieval.
- **Liu et al., *Lost in the Middle*.** Why a bigger context window is not a substitute for good retrieval.
- **Halliday & Hasan, *Cohesion in English* and Toulmin, *The Uses of Argument*.** The linguistics and logic behind what chunking destroys.
- **Microsoft Research, *GraphRAG*.** A glimpse of the road past chunking: structured, relational retrieval.

- **Trivedi et al., *Interleaving Retrieval with Chain-of-Thought*.** Why single-pass chunking often fails for multi-hop reasoning.

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